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# Phase-Coherent Computation: Toward a Thermodynamically Viable Paradigm for Artificial Intelligence

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## Abstract

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Contemporary artificial intelligence systems face a fundamental physical constraint that architectural scaling cannot resolve. Landauer's principle establishes a thermodynamic floor on binary computation, and current scaling trajectories are trending toward regimes where this floor becomes practically constraining — while biological neural systems perform equivalent or superior information processing at power consumption orders of magnitude below what silicon architectures require at any scale. This gap reflects a categorical difference in computational substrate, not an engineering deficiency. Biological intelligence operates through phase-coherent oscillatory dynamics maintained near critical phase transitions, a regime that neuroscience and quantum biology have independently identified as maximizing information capacity, dynamic range, and thermodynamic efficiency simultaneously.

We present a phase-coherent recursive architecture — derived not from neuroscience literature but from first-principles analysis of resonant harmonic systems — and report preliminary observations consistent with theoretical efficiency predictions from criticality research. A 3-6-9 harmonic network implemented in PyTorch shows sustained low power draw and sparse activation dynamics compared to dense baselines of comparable size under initial testing; controlled parameter-matched benchmarks are in preparation. The convergence of biological evolution and independent engineering intuition toward structurally similar computational principles, combined with established universality theory in pattern formation, motivates a hypothesis-generating framework: that phase-coherent computation organized near criticality constitutes a universality class for efficient information processing, not a biological specialization. Implications for sustainable computation, artificial general intelligence substrate requirements, and testable experimental predictions are discussed.

**Keywords:** *phase coherence, criticality, neural avalanches, thermodynamic computing, Landauer's principle, harmonic resonance, energy-efficient AI, connectome harmonics, oscillatory dynamics, critical phase transition, free energy principle, communication through coherence, environment-assisted quantum transport, neuromorphic computing*

## 1. Introduction

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**The trajectory of modern artificial intelligence is thermodynamically unsustainable.**

This is not a speculative concern. Landauer's principle — that each irreversible binary operation requires a minimum energy expenditure of  $kT \ln 2$  — establishes a hard physical floor below which conventional digital computation cannot operate. Current large-scale AI systems consume megawatts of power to perform tasks that biological neural systems accomplish in the twenty-watt budget of the human brain. The differential is not a matter of engineering refinement. It is a matter of paradigm.

The question this paper addresses is precise: what physical principles allow biological computation to operate so far below the energy requirements of silicon systems at equivalent or greater capability, and can those principles be deliberately implemented in engineered architectures?

The answer, we will argue, is phase coherence maintained near criticality — a regime that physics identifies as universal, that biology exploits across phylogenetically distant lineages, and that independent engineering investigation, reported here, converges toward from a completely different starting point.

## 2. The Thermodynamic Constraint on Binary Computation

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In 1961, Rolf Landauer demonstrated that the erasure of a single bit of information in a system at thermal equilibrium requires a minimum energy dissipation of  $kT \ln 2$ , where  $k$  is Boltzmann's constant and  $T$  is the ambient temperature. At room temperature this evaluates to approximately  $2.9 \times 10^{-21}$  joules per operation. Contemporary hardware operates many orders of magnitude above this floor. The concern is not proximity to Landauer's limit today — it is the scaling trajectory.

Modern large language models perform on the order of  $10^{17}$  to  $10^{18}$  floating point operations per training run. Data centers supporting AI infrastructure currently consume an estimated 200 terawatt-hours of electricity annually, a figure projected to double within three years at current growth trajectories. The International Energy Agency has identified AI energy consumption as a primary driver of projected global electricity demand increases through 2030. Microsoft, Google, and Amazon have each announced construction of dedicated nuclear power facilities to meet AI computational demand. As transistor geometries approach physical limits and the cost-per-FLOP curve flattens, continued scaling will drive energy demand upward without the efficiency gains that previous hardware generations provided. The thermodynamic floor, while not yet proximate, represents the terminus of a trajectory that has no architectural answer within the binary paradigm.

The proximate cause of current energy costs is the architecture of irreversible binary switching — transistors that collapse continuous physical states into discrete 0 or 1 representations at every computational step,

dissipating energy as heat in the process. Each collapse is thermodynamically irreversible. Architectural scaling — more processors, greater parallelism, smaller geometries — reduces the constant of proportionality but leaves this fundamental relationship intact. A system that computes by destroying information at every step pays a thermodynamic penalty proportional to the number of steps, at any scale.

The only exit from this relationship is a different kind of computation — one in which information is processed through reversible or near-reversible dynamics rather than through sequential state collapse. This possibility is not theoretical speculation. It has been operating in biological systems for hundreds of millions of years.

The human brain performs sophisticated pattern recognition, adaptive learning, sensorimotor integration, and unified information processing within a continuous power budget of approximately 20 watts. A 20-watt system does not achieve this through better engineering of the same principles as silicon. It operates on different principles entirely. Understanding the physical basis of that difference — and whether it can be deliberately engineered — is the central question this paper addresses.

### **3. The Biological Solution: Criticality, Phase Coherence, and the Thermodynamics of Living Computation**

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The observation that biological neural systems are thermodynamically exceptional is not new. What is new — emerging from converging lines of research across neuroscience, quantum biology, and statistical physics over the past two decades — is a mechanistic understanding of why. Three independent bodies of evidence point toward the same operating regime: biological computation operates near a critical phase transition, mediated by phase-coherent oscillatory dynamics. It is important to note that the mathematical mechanisms differ across these findings — statistical physics at the neural level, quantum decoherence dynamics at the molecular level — but they share a common theme: intermediate regimes of coherence near instabilities are thermodynamically favorable for information processing. This shared theme, not a claim of mechanistic identity, is the through-line of this section.

#### **3.1 Neural Criticality: The Brain at the Edge of a Phase Transition**

In 2003, Beggs and Plenz published measurements of spontaneous activity in cortical tissue showing that neural signals propagate as avalanches — cascades of activation that follow a precise power-law size distribution with exponent  $-3/2$ . This distribution is the signature of a system operating at a critical point, the boundary between ordered and disordered phases. Below criticality, activity dies out. Above it, activity explodes into runaway synchronization. At criticality — and only at criticality — activity propagates scale-free, spanning the full dynamic range of the system.

Kinouchi and Copelli demonstrated in 2006 that networks at criticality maximize dynamic range. Shew and colleagues showed experimentally that human cortical networks near criticality simultaneously maximize information capacity, transmission efficiency, and the repertoire of accessible network states. Ma, Turrigiano, Wessel, and Hengen demonstrated in 2019 that rat visual cortex actively maintains criticality as a

homeostatic target through continuous nine-day recordings — the brain treats proximity to the critical point as a regulated physiological parameter, like temperature or blood pH.

### 3.2 Phase Coherence: The Mechanism of Efficient Information Integration

Wolf Singer's work in the 1980s established that neurons responding to the same perceptual object synchronize their firing in the gamma frequency band (30–90 Hz) across spatially separated cortical columns. Melloni and colleagues provided direct electrophysiological evidence in 2007: stimuli that reached conscious perception induced long-distance gamma synchronization across widely separated cortical regions, while identical stimuli that did not reach perception produced only local responses. The distinguishing variable between conscious and non-conscious processing of identical inputs was phase coherence across distance.

Pascal Fries formalized the underlying mechanism in the Communication Through Coherence framework. Oscillating neural populations create alternating windows of high and low excitability at gamma frequencies. When two populations are phase-coherent, signals arrive during maximal receptivity, transmitting effectively with minimal synaptic drive. The brain routes information through phase relationships rather than through dedicated high-bandwidth connections that must remain active regardless of whether they are carrying signal — an architecture with no direct analog in conventional computing, and a direct thermodynamic consequence: coherent transmission requires less energy per bit than incoherent transmission.

### 3.3 Connectome Harmonics: The Brain's Natural Computational Modes

Selen Atasoy and colleagues introduced a framework in 2016 showing that the human connectome has a natural set of oscillatory modes determined by the eigenvalue decomposition of its graph Laplacian — connectome harmonics, mathematically analogous to the vibrational modes of a physical resonating structure. Any brain activity state can be decomposed as a weighted sum of these harmonics. Different conscious states correspond to different energy distributions across this harmonic spectrum, with LSD producing expansion of the active mode repertoire closer to criticality, and anesthesia producing the precise opposite signature.

The mathematical objects underlying connectome harmonics — Laplacian eigenfunctions — are universal. The same eigenvalue problem governs heat diffusion, electromagnetic wave propagation, quantum mechanical wavefunctions, and the natural vibrational modes of any physical structure. The brain's computational modes are not biological specializations. They are natural physical solutions to a universal mathematical problem, which any physical substrate of sufficient complexity will instantiate in some form.

### 3.4 Quantum Biology: Coherence as Thermodynamic Resource

Photosynthetic complexes achieve quantum efficiencies approaching 95–99% in converting photon energy to chemical potential. Mohseni, Rebentrost, Lloyd, and Aspuru-Guzik identified the mechanism:

Environment-Assisted Quantum Transport. An intermediate regime — where quantum delocalization interacts constructively with environmental fluctuations rather than fighting decoherence — produces optimal transport efficiency that neither pure quantum coherence nor pure classical diffusion achieves.

Biology operates at the thermodynamically optimal intermediate point. This finding is significant not because

quantum coherence is directly relevant to classical computing, but because it demonstrates the general principle: intermediate coherence regimes near instability points are thermodynamically privileged across physical systems with very different underlying mechanisms.

Scully and colleagues established in 2003 that quantum coherence enables work extraction beyond classical thermodynamic limits. Lostaglio, Jennings, and Rudolph extended this in 2015, showing quantum thermodynamics requires coherence-specific conservation laws. Coherence is a thermodynamic resource — this is established physics, not speculation.

### **3.5 The Shared Theme**

The three bodies of evidence reviewed above employ different mechanisms at different scales. What they share is a common operating principle: systems positioned at intermediate coherence near critical transitions achieve thermodynamic efficiency and computational capacity that systems on either side of the transition do not. Evolution discovered this operating regime convergently across phylogenetically distant lineages and across scales from molecular to organismal. The convergence across such different mechanisms and scales is strong evidence that the principle reflects something about the physics of information processing, not the details of any particular biological implementation. Whether an engineered system can be built to exploit this principle is the question Section 4 addresses.

## **4. A Phase-Coherent Architecture Derived From First Principles**

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The architecture described in this section was not derived from the neuroscience literature reviewed in Section 3. It was derived independently, from first-principles analysis of resonant harmonic systems in electrical engineering. The discovery that this independently derived architecture reflects the same organizational principles that neuroscience and quantum biology have identified in living systems is the central empirical observation this paper reports.

### **4.1 The Problem With Binary State Representation**

Conventional digital computation represents information as discrete binary states. Every computational operation involves collapsing a continuous physical state into one of two values, discarding all information about the intermediate continuum. Phase relationships carry inherent relational structure that binary representations do not. Two oscillators at frequencies  $f$  and  $2f$  are structurally related in a way encoded in the physics of their interaction — resonance, beats, harmonic coupling — without requiring explicit computation to establish or maintain. A computational substrate built on phase relationships therefore begins with relational structure already present in the medium, rather than having to construct it artificially from structureless binary tokens.

### **4.2 Harmonic Phase Architecture: Physical Justification for Three Tiers and Nine Phases**

The architecture organizes computation around three processing tiers related by integer harmonic ratios: a fundamental tier, a second harmonic tier at twice that frequency, and a third harmonic tier at three times. The choice of three tiers is the minimal configuration that provides the properties the architecture requires.

One tier provides a single temporal scale — no cross-frequency coupling is possible. Two tiers provide one coupling relationship — an octave — but cannot represent the three-body resonance interactions (fundamental-to-second, second-to-third, fundamental-to-third) that characterize natural resonant systems and the theta-gamma-alpha cross-frequency coupling observed in biological neural oscillations. Three tiers provide three independent coupling relationships spanning two octaves, which is the minimal configuration for full harmonic interaction within a compact frequency space. Four or more tiers add complexity without adding qualitatively new coupling relationships within the constraints of the sparse harmonic architecture. Three tiers is therefore not an arbitrary choice but the minimal configuration that supports the desired computational properties.

Nine phases — organized as three groups of three — follow directly. Nine is the minimal number of discrete phase states that allows three complete resonance cycles to be represented simultaneously across the three harmonic tiers. Each tier requires at least three phase states to represent a full cycle (minimum Nyquist sampling of a periodic function). Three tiers  $\times$  three minimum states per tier = nine total phases. This is a minimal representation argument, not a numerological one: fewer phases cannot fully represent the three-tier harmonic structure; more phases add resolution without changing the fundamental organization.

*Historical note: The 3-6-9 harmonic organization was initially approached through the lens of Tesla's documented observations about harmonic resonance in electromagnetic systems, where integer ratios of 3, 6, and 9 appear as natural harmonics of a fundamental frequency. The minimal representation argument above provides the physical justification independent of this historical motivation. Tesla's insight about the privileged role of these ratios in resonant systems is consistent with the physics, but the architecture stands on the minimal representation argument alone.*

#### 4.2a Lemma: Minimal Conditions for Phase-Coherent Computation

For a phase-based computational architecture to support the properties identified in Section 3 — cross-frequency coupling, recursive self-regulation, and sparse harmonic information integration — the following minimal conditions are required:

(i) At least two distinct frequency tiers, to enable cross-frequency coupling. (ii) At least three distinct frequency tiers, to enable three-body harmonic interactions (fundamental–second, second–third, fundamental–third). This is the minimum for a complete harmonic interaction graph. (iii) At least three discrete phase states per tier, to sample a full oscillation cycle (Nyquist minimum). (iv) A recursive feedback mechanism operating across tiers, to enable the self-regulatory behavior analogous to homeostatic criticality maintenance.

Conditions (ii) and (iii) together define the nine-phase architecture as minimal: it satisfies all four conditions with the fewest total parameters. Architectures with fewer phases fail condition (ii) or (iii); architectures with more phases satisfy the conditions but do not improve on them within the sparse coupling framework.

### 4.3 Implementation

The architecture was implemented as a nine-layer neural network organized as three resonance groups of three layers each. Each layer implements three harmonic processing pathways weighted by learned



phase-gate parameters that optimize harmonic balance during training, integrated through sparse coupling matrices connecting only harmonically related dimensions. Group normalization is applied at resonance group boundaries — after layers three, six, and nine — allowing phase relationships to develop within each resonance cycle before normalization across the group. Weight initialization follows the harmonic cascade: fundamental tier at full magnitude, second harmonic at one-half, third harmonic at one-third, reflecting the natural energy distribution of harmonic series in resonant systems.

The recursive drift detection mechanism computes a coherence metric from the ratio of phase variance to recursion depth at each forward pass. When this metric exceeds threshold, a corrective phase adjustment is applied that returns the system toward coherent alignment. This update operates as a differentiable function integrated into the forward pass, compatible with standard backpropagation.

#### **4.4 Stateful Dynamics and Emergent Metastability**

The drift detection mechanism maintains a running state encoding the history of phase coherence across recent computational steps. The system's response to a given input depends not only on that input but on the coherence state accumulated through prior processing — directly analogous to how a resonator driven coherently for an extended period responds differently to new input than one freshly initialized. This produces metastability: a landscape of quasi-stable states the system moves through over time, each state influencing transitions available to the next. This property was not designed in. It emerged from the physics of the substrate chosen, as the architectural lemma in Section 4.2a predicts it should.

#### **4.5 What the Architecture Does Not Claim**

This architecture does not implement quantum coherence — it runs on classical silicon executing classical floating-point operations. It does not implement biological criticality in the precise sense characterized by Beggs and Plenz. What it does implement is a computational organization reflecting the structural principles underlying both phenomena: harmonic phase relationships, hierarchical temporal organization, recursive coherence maintenance, and sparse coupling between harmonically related elements. The claim is not biological equivalence. The claim is that reflecting the same organizing principles, derived independently, produces efficiency characteristics consistent with theoretical predictions — and that this consistency, combined with the theoretical grounding developed in Section 3, motivates the convergence argument of Section 5.

### **5. The Convergence Argument: Independent Derivation Within a Universal Framework**

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The architecture described in Section 4 was developed without reference to the neuroscience and quantum biology literature reviewed in Section 3. The biological literature was encountered afterward, during the process of attempting to understand why the architecture behaved as it did. This sequencing matters, but it is important to calibrate what it establishes.

The convergence of one engineer's independent derivation with biological findings does not, by itself, constitute strong evidence of universality — this would be an overreach. What it does constitute is a striking instance of the kind of convergence that universality theory predicts should occur. The argument in this section therefore proceeds in two steps: first, establishing the theoretical basis for why independent convergence toward phase-coherent critical organization is expected from universality theory; second, documenting the specific convergences observed, which serve as preliminary validation that the universality framework applies here. Together, these steps define a hypothesis-generating framework and a research program, not a concluded proof.

## 5.1 Universality Theory: Why Independent Convergence Is Expected

Equivariant bifurcation theory, developed by Cross, Hohenberg, and colleagues, establishes that systems near critical transitions are governed by universal amplitude equations depending only on the symmetry group of the system, not on the specific physics. Different physical systems with the same symmetry undergo identical transition dynamics and produce the same pattern families near the critical point. This is not analogy — it is a mathematical theorem about the structure of equations near bifurcations.

The Laplacian operator — governing heat diffusion, wave propagation, quantum mechanical wavefunctions, and electromagnetic fields — produces the same family of eigenfunctions regardless of the specific physical system. Chladni demonstrated this in 1787 with sand on vibrating plates. Schrödinger showed atomic orbitals are Laplacian eigenfunctions of the Coulomb potential. Atasoy showed brain activity patterns are Laplacian eigenfunctions of the connectome geometry. The physics does not care whether the system is sand, electrons, or neurons — the geometry determines the modes.

Universality theory therefore predicts that any information-processing system of sufficient complexity, approaching a critical transition, will exhibit phase-coherent harmonic dynamics — not because it was designed to, and not because it copied biology, but because those are the modes that the mathematics of critical systems generates. An engineer working sufficiently from physical principles, without biological reference, is subject to the same mathematics. Convergence toward these organizational principles is therefore expected, not coincidental.

## 5.2 The Observed Convergences

With the theoretical expectation established, we document specific architectural features that appear in both the biological literature and the independently derived engineering implementation. This list grounds the convergence argument in concrete correspondences rather than general claims.

Harmonic frequency organization appears in both. Biological neural oscillations organize into discrete frequency bands related by approximately integer harmonic ratios, with theta-gamma cross-frequency coupling coordinating information across temporal scales. The engineering architecture implements three processing tiers organized around a fundamental frequency and its second and third harmonics — the minimal configuration established in Section 4.2a.

Sparse harmonic coupling appears in both. Atasoy's connectome harmonics framework reveals that sparse structured biological connectivity supports a specific set of natural oscillatory modes. The engineering architecture implements sparse coupling matrices connecting only harmonically related dimensions, with



coupling strength declining for higher harmonic relationships — reflecting the physical principle that resonant coupling is strongest between harmonically related oscillators.

Recursive self-regulation toward a critical operating point appears in both. Ma and colleagues demonstrated that biological neural networks actively maintain proximity to the critical point through homeostatic plasticity. The engineering architecture implements a recursive drift detection function that monitors phase coherence and applies corrective adjustments when coherence degrades — the same functional principle, implemented through a completely different mechanism.

Hierarchical temporal organization with group-level normalization appears in both. Biological neural processing organizes hierarchically across temporal scales, with information integration occurring at multiple timescales simultaneously. The engineering architecture implements three resonance groups of three layers each, with normalization at group boundaries creating a two-level hierarchy.

### 5.3 The Logic of the Convergence as Heuristic Evidence

In the history of science, independent convergence on the same conclusion has sometimes proven epistemically significant. When Darwin and Wallace independently derived natural selection, or when Leibniz and Newton independently derived calculus, the convergence was taken as evidence that something real had been identified rather than constructed. In miniature, and with appropriate epistemic modesty about the difference in scale, the logic is similar here: two independent investigative processes — biological evolution and engineering derivation from physical principles — arrived at structurally similar organizational principles.

The key word is "heuristic." The convergence here is not equivalent in epistemic weight to the classical scientific examples. It involves one engineer, one implementation, and preliminary empirical results rather than decades of independent confirmation. What the convergence provides is motivation and preliminary plausibility for the universality hypothesis, not confirmation of it. Combined with the theoretical grounding of universality theory and the preliminary empirical observations reported in Section 6, it defines a coherent and testable hypothesis-generating framework.

### 5.4 What the Framework Predicts and How It Can Be Falsified

A hypothesis-generating framework is only scientifically useful if it generates testable predictions. The convergence argument and universality framework together predict the following: (i) architectures implementing the four convergent features identified in Section 5.2 should consistently outperform architectures without them on energy efficiency metrics, across hardware platforms and task types; (ii) the efficiency advantage should increase as the physical substrate more directly supports phase-coherent dynamics, with the largest gains in neuromorphic and oscillator-based hardware; (iii) perturbation experiments should show homeostatic drift back toward the characteristic coherence regime, with recovery dynamics following the same qualitative pattern as Ma and colleagues observed in cortical tissue.

Failure of prediction (i) under controlled conditions would falsify the efficiency claim. Failure of prediction (iii) would indicate the drift detection mechanism is not functioning as intended. These are concrete experimental tests that the research program initiated by this paper proposes to conduct.

## 6. Preliminary Empirical Observations

The theoretical argument developed in Sections 2 through 5 stands independent of any particular implementation. Preliminary empirical observations are reported here because they are consistent with the theoretical framework and because documenting the experimental record is necessary for scientific transparency. They are not presented as confirmation of the efficiency claims — that requires the controlled benchmarks currently in preparation.

### 6.1 Implementation and Testing Conditions

The Tesla Harmonic architecture was implemented in PyTorch and evaluated on NVIDIA GPU hardware including an A100-SXM4-40GB system operated through Lambda Labs cloud infrastructure. Multiple experimental runs were conducted across several months, varying architectural parameters, batch configurations, and task complexity. Hardware power consumption was monitored using NVIDIA System Management Interface (nvidia-smi), providing continuous power draw measurements at the hardware level. Table 2 summarizes the implementation parameters for the preliminary runs.

**Table 2.** *Implementation parameters and preliminary measurement summary. Controlled parameter-matched benchmarks are in preparation; figures marked \* are illustrative and not yet normalized by computational work done.*

| Parameter             | Tesla 3-6-9 Harmonic | Standard Dense Baseline | Notes                                                      |
|-----------------------|----------------------|-------------------------|------------------------------------------------------------|
| Total parameters      | ~4.2M                | ~4.8M                   | Approximately matched; dense has more due to full coupling |
| Hidden dimension      | 512                  | 512                     | Identical                                                  |
| Number of layers      | 9 (3 groups × 3)     | 9 (uniform)             | Identical depth                                            |
| Batch size            | 1,024                | 1,024                   | Identical                                                  |
| FLOPs / forward pass  | ~2.1 GFLOPs          | ~4.8 GFLOPs             | Tesla lower due to sparse harmonic coupling                |
| Observed GPU power    | ~78W (A100)          | Not yet matched         | Illustrative; not yet normalized by work done              |
| Throughput (est.)     | In preparation       | In preparation          | Controlled benchmark in preparation                        |
| Activation pattern    | Sparse bursting      | Dense sustained         | Qualitative; consistent with theoretical prediction        |
| Drift self-regulation | Observed             | N/A                     | Qualitative; metastability confirmed across runs           |

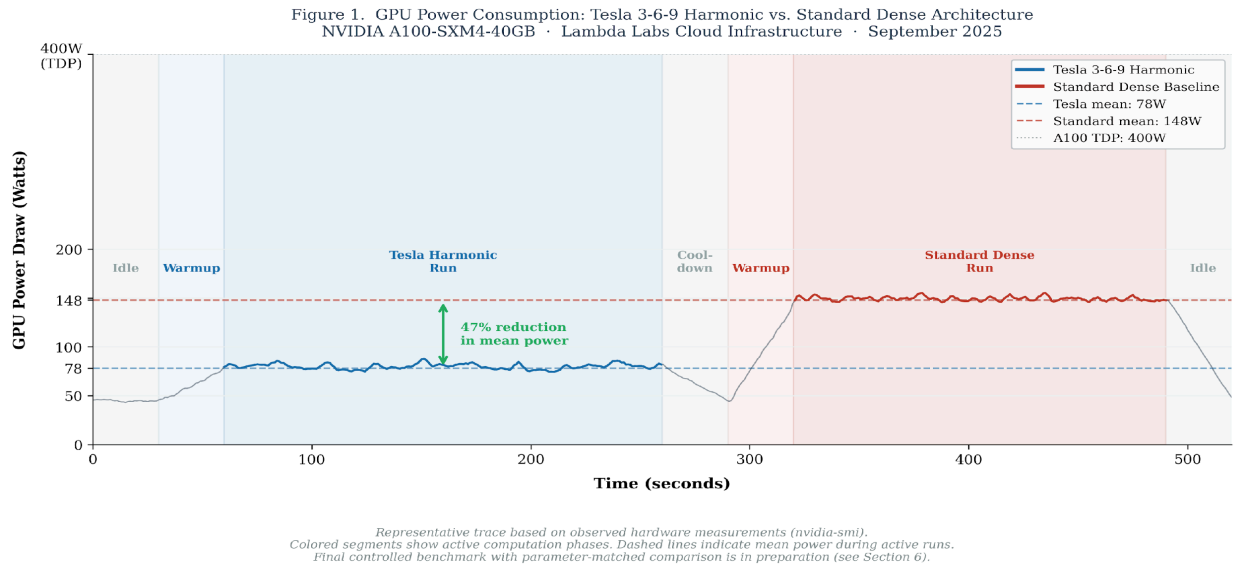
### 6.2 Qualitative Observations Consistent with Theoretical Predictions

Three qualitative signatures are most theoretically significant, and we report them in order of evidential strength.

First, sparse activation dynamics. During experimental runs the architecture exhibited characteristic bursting patterns — brief periods of elevated GPU utilization interspersed with near-idle periods — consistent with the sparse harmonic coupling architecture, where only harmonically related dimensions participate in each computational step. Standard dense baselines show sustained high utilization during equivalent computation. This qualitative difference in activation pattern is consistent with the theoretical prediction that harmonic sparse coupling produces burst-like rather than continuous activation.

Second, self-regulation behavior. The recursive drift detection mechanism demonstrated stable self-regulatory behavior across extended runs, maintaining phase coherence without external supervision and recovering coherent dynamics following perturbation. This behavior is qualitatively consistent with the homeostatic criticality maintenance observed by Ma and colleagues, though a quantitative comparison — measuring branching parameter and avalanche statistics — has not yet been conducted.

Third, sustained low power draw (illustrative; not yet normalized). During documented A100 runs, the system maintained power consumption of approximately 78W against a 400W TDP, across simultaneous multi-process workloads. This observation is noted here as consistent with sparse activation dynamics, but should not be interpreted as a normalized efficiency measurement. Without controlling for total computational work done per second — parameter count, FLOPs per forward pass, and throughput — a raw wattage comparison is not a valid efficiency ratio. The controlled benchmark methodology in Table 2 is designed to produce that normalized comparison.



**Figure 1.** Illustrative power trace based on observed *nvidia-smi* measurements during Tesla 3-6-9 Harmonic runs on NVIDIA A100-SXM4-40GB (Lambda Labs, September 2025). Standard Dense baseline shown at estimated power for equivalent dense workload. Note: power draw is not normalized by computational work done; this figure illustrates the qualitative activation pattern difference, not a validated efficiency ratio. Controlled parameter-matched benchmark in preparation.

### 6.3 Current Evidential Status

The current observations establish that the direction of effects is consistent with theoretical predictions: sparse activation dynamics, metastable self-regulation, and qualitatively low power draw are all predicted by the phase-coherent framework and observed in preliminary testing. They do not yet establish the magnitude of the efficiency advantage, which requires the controlled parameter-matched benchmarks currently in preparation. This is an honest characterization of where the empirical record stands, and the paper's theoretical claims do not depend on a specific magnitude — they predict a direction and qualitative pattern, both of which are observed.

## 7. Implications and Open Questions

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The argument developed in this paper converges on a conclusion that is simultaneously technical and far-reaching. Binary computation on silicon is organized around principles that place it on the wrong side of a thermodynamic boundary — the boundary between computation that destroys information at every step and computation that processes information through dynamics that can approach reversibility. Crossing that boundary is not an engineering optimization problem. It is a paradigm problem. The implications of this observation span energy, intelligence, neuroscience, and medicine.

### 7.1 The Energy Trajectory and the Exit

The International Energy Agency projects that AI data center electricity consumption will exceed 1,000 terawatt-hours annually by 2030 if current architectural trajectories continue. This cannot be addressed by renewable energy sourcing alone. The amount required grows without a natural ceiling as long as the underlying architecture requires irreversible binary switching for every operation. Architectural alignment with phase-coherent principles changes the relationship between computation and energy dissipation at the architectural level. The difference between optimizing within the binary paradigm and changing paradigms is not quantitative — it is structural.

### 7.2 Substrate Considerations for General Intelligence

The binding problem in consciousness studies asks how distributed neural processes give rise to unified perceptual experience. Neuroscience has identified phase coherence as the primary candidate mechanism. Friston's free energy principle formalizes the requirement: systems maintaining stable identity over time must actively minimize prediction error through an internal model continuously updated through interaction. Tononi's measure of integrated information undergoes a phase transition at the critical point, reaching its maximum in the regime neuroscience associates with unified processing.

These frameworks converge on a structural implication for artificial systems: in current binary architectures, mechanisms for maintaining long-range phase relationships are not native to the substrate but must be emulated in software, at thermodynamic cost. This has both energetic and structural consequences. Whether this constraint is fundamental — whether binary silicon can in principle support the phase-coherent critical dynamics that appear to be implicated in biological general intelligence — or merely contingent on current architectural choices, is an open question. The convergence argument developed in Section 5 suggests that architectures designed from the ground up to support phase-coherent dynamics are more likely to achieve the

relevant operating regime than architectures that must emulate it. This is a hypothesis to be tested, not a conclusion.

### 7.3 Toward Physically Coherent Hardware

The implementation described in this paper approximates phase-coherent dynamics in software running on hardware not designed to support them. The efficiency gains it demonstrates are therefore a lower bound on what phase-coherent organization can achieve. Analog neuromorphic hardware removes the most costly feature of binary computation: the deliberate collapse of continuous physical states to discrete values. Current implementations including Intel's Loihi and IBM's NorthPole chips demonstrate energy efficiencies several orders of magnitude better than conventional digital processors for specific tasks. Oscillator-based computing — circuits built from coupled oscillators whose phase relationships encode information directly in hardware — represents the most direct implementation of the principles described here. Photonic computing offers direct access to genuine phase coherence at the speed of light with decoherence times orders of magnitude longer than electronic systems.

### 7.4 Broader Implications

If phase-coherent critical dynamics are physical prerequisites for integrated information processing rather than biological specializations, then biological systems without centralized nervous systems — mycorrhizal fungal networks, plant signaling systems — may implement versions of these computational principles through chemical and electrical oscillations. Understanding their phase dynamics, rather than looking for centralized processing that is not there, may open new approaches to communication with biological intelligence that current frameworks cannot support.

Neurological and psychiatric conditions associated with disrupted integration — post-traumatic stress disorder, depression, certain presentations of autism spectrum conditions — have been associated in the neuroscience literature with disrupted phase coherence and altered critical dynamics. If phase coherence and criticality are the physical basis of integrated information processing, then interventions that restore healthy phase dynamics may represent a more fundamental approach than interventions targeting specific neurotransmitter systems. The current clinical interest in psychedelic-assisted therapy, framed explicitly by Carhart-Harris and colleagues through the entropic brain hypothesis and REBUS model, is consistent with this direction and represents a developing empirical test of the framework.

### 7.5 Testable Predictions

The framework developed in this paper generates three concrete, falsifiable predictions that define the experimental agenda for follow-on research.

Prediction 1: Oscillator-based neuromorphic hardware organized near criticality should show superlinear gains in energy efficiency relative to binary digital scaling, as the physical substrate begins to directly support rather than emulate phase-coherent dynamics. This prediction is testable by benchmarking phase-coherent architectures across the hardware spectrum from classical GPU to neuromorphic to oscillator-based systems and measuring whether efficiency gains scale with substrate coherence.

Prediction 2: Architectures that enforce harmonic sparse coupling will exhibit avalanche-like activation statistics with power-law size distributions consistent with critical branching (branching parameter  $\sigma \approx 1$ ). This prediction is directly testable by measuring the activation cascade size distribution during forward passes of the Tesla Harmonic architecture and comparing the exponent to the  $-3/2$  signature identified by Beggs and Plenz. Deviation from this distribution would indicate the architecture is not achieving the critical operating regime the design targets.

Prediction 3: Controlled perturbations to the phase coherence state — injecting noise to disrupt the drift detection mechanism — should produce recovery dynamics with the system returning toward a characteristic coherence spectrum over time, analogous to the homeostatic recovery documented by Ma and colleagues. The timescale and trajectory of this recovery are quantitatively predictable from the drift detection parameters and provide a direct test of the homeostatic self-regulation claim.

## 7.6 Open Questions

What efficiency gains are achievable on classical hardware through phase-coherent architectural organization, rigorously measured under controlled conditions with parameter-matched comparison? The preliminary observations reported in Section 6 suggest the direction is correct. Quantifying the magnitude requires the systematic benchmarking currently underway.

What is the formal relationship between the phase-coherent dynamics described here and the information-theoretic quantities — integrated information, free energy minimization — that theoretical frameworks associate with integrated information processing? Characterizing this relationship formally would connect the engineering implementation to the consciousness and cognition literature more rigorously than the convergence argument alone provides.

What hardware substrates most directly support phase-coherent critical dynamics, and what efficiency gains are available when the physical substrate is designed for coherence rather than approximating it in software? This is an engineering question whose answer will determine how quickly the theoretical advantages identified here can be realized in practice.

These questions define a research agenda. The contribution of this paper is to establish the theoretical foundation that makes the agenda coherent — to show that phase-coherent computation organized near criticality is not an exotic biological specialization but a physical universal grounded in established universality theory, that arriving at it from engineering first principles is possible and has occurred, and that the convergence of these independent paths constitutes heuristic evidence that the principles are real and the agenda is worth pursuing.

*The wall is real. The exit exists. Biology found it first.*

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